

9. CRUD Operations using MongoDB

Design and implement basic Create, Read, Update, and Delete (CRUD) operations using MongoDB. Use the save method and logical operators where necessary

gescoe@gescoe-OptiPlex-3010:~\$ mongosh

Current Mongosh Log ID: 69eef2af0eb7daf5fce5f46

Connecting to:

mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000&ap
pName=mongosh+2.5.8

Using MongoDB: 8.2.1

Using Mongosh: 2.5.8

mongosh 2.5.9 is available for download: <https://www.mongodb.com/try/download/shell>

For mongosh info see: <https://www.mongodb.com/docs/mongodb-shell/>

The server generated these startup warnings when booting

2026-04-27T10:33:34.149+05:30: Using the XFS filesystem is strongly recommended with the WiredTiger storage engine. See <http://dochub.mongodb.org/core/prodnotes-filesystem>

2026-04-27T10:33:41.826+05:30: Access control is not enabled for the database. Read and write access to data and configuration is unrestricted

2026-04-27T10:33:41.827+05:30: For customers running the current memory allocator, we suggest changing the contents of the following sysfsFile

2026-04-27T10:33:41.827+05:30: For customers running the current memory allocator, we suggest changing the contents of the following sysfsFile

2026-04-27T10:33:41.827+05:30: We suggest setting the contents of sysfsFile to 0.

2026-04-27T10:33:41.827+05:30: We suggest setting swappiness to 0 or 1, as swapping can cause performance problems.

test> use Stud;

switched to db Stud

Stud> db.createCollection("StudInfo");

{ ok: 1 }

Stud> db.Stud.insertOne({Roll_No:1,Name:"Gita",Address:"Nashik"});

```
{
  acknowledged: true,
  insertedId: ObjectId('69eef916f0eb7daf5fce5f47')
}
```

Stud> db.Stud.find();

```
[
  {
    _id: ObjectId('69eef916f0eb7daf5fce5f47'),
    Roll_No: 1,
    Name: 'Gita',
    Address: 'Nashik'
  }
]
```

```

Stud>
db.Stud.insertMany([ {Roll_No:2,Name:"Sita",Address:"Pune"}, {Roll_No:3,Name:"Ram",Address:
"Pune"}, {Roll_No:4,Name:"Sham",Address:"Nashik"} ]);
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69eeb2bf0eb7daf5fce5f48'),
    '1': ObjectId('69eeb2bf0eb7daf5fce5f49'),
    '2': ObjectId('69eeb2bf0eb7daf5fce5f4a')
  }
}
Stud> db.Stud.find();
[
  {
    _id: ObjectId('69eef916f0eb7daf5fce5f47'),
    Roll_No: 1,
    Name: 'Gita',
    Address: 'Nashik'
  },
  {
    _id: ObjectId('69eeb2bf0eb7daf5fce5f48'),
    Roll_No: 2,
    Name: 'Sita',
    Address: 'Pune'
  },
  {
    _id: ObjectId('69eeb2bf0eb7daf5fce5f49'),
    Roll_No: 3,
    Name: 'Ram',
    Address: 'Pune'
  },
  {
    _id: ObjectId('69eeb2bf0eb7daf5fce5f4a'),
    Roll_No: 4,
    Name: 'Sham',
    Address: 'Nashik'
  }
]

```

```

Stud> db.Stud.updateOne({Roll_No:1},{ $set: {Roll_No:5}});
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
Stud> db.Stud.find();
[
  {
    _id: ObjectId('69eef916f0eb7daf5fce5f47'),
    Roll_No: 5,

```

```

    Name: 'Gita',
    Address: 'Nashik'
  },
  {
    _id: ObjectId('69eeb2bf0eb7daf5fce5f48'),
    Roll_No: 2,
    Name: 'Sita',
    Address: 'Pune'
  },
  {
    _id: ObjectId('69eeb2bf0eb7daf5fce5f49'),
    Roll_No: 3,
    Name: 'Ram',
    Address: 'Pune'
  },
  {
    _id: ObjectId('69eeb2bf0eb7daf5fce5f4a'),
    Roll_No: 4,
    Name: 'Sham',
    Address: 'Nashik'
  }
]

```

```

Stud> db.Stud.find().sort({Roll_No: -1});
[
  {
    _id: ObjectId('69eef916f0eb7daf5fce5f47'),
    Roll_No: 5,
    Name: 'Gita',
    Address: 'Nashik'
  },
  {
    _id: ObjectId('69eeb2bf0eb7daf5fce5f4a'),
    Roll_No: 4,
    Name: 'Sham',
    Address: 'Nashik'
  },
  {
    _id: ObjectId('69eeb2bf0eb7daf5fce5f49'),
    Roll_No: 3,
    Name: 'Ram',
    Address: 'Pune'
  },
  {
    _id: ObjectId('69eeb2bf0eb7daf5fce5f48'),
    Roll_No: 2,
    Name: 'Sita',
    Address: 'Pune'
  }
]
Stud> db.Stud.find().sort({Roll_No: 1});
[

```

```

{
  _id: ObjectId('69eeb2bf0eb7daf5fce5f48'),
  Roll_No: 2,
  Name: 'Sita',
  Address: 'Pune'
},
{
  _id: ObjectId('69eeb2bf0eb7daf5fce5f49'),
  Roll_No: 3,
  Name: 'Ram',
  Address: 'Pune'
},
{
  _id: ObjectId('69eeb2bf0eb7daf5fce5f4a'),
  Roll_No: 4,
  Name: 'Sham',
  Address: 'Nashik'
},
{
  _id: ObjectId('69eef916f0eb7daf5fce5f47'),
  Roll_No: 5,
  Name: 'Gita',
  Address: 'Nashik'
}
]

```

```

Stud> db.Stud.deleteOne({Roll_No:5});
{ acknowledged: true, deletedCount: 1 }
Stud> db.Stud.find();

```

```

[
  {
    _id: ObjectId('69eeb2bf0eb7daf5fce5f48'),
    Roll_No: 2,
    Name: 'Sita',
    Address: 'Pune'
  },
  {
    _id: ObjectId('69eeb2bf0eb7daf5fce5f49'),
    Roll_No: 3,
    Name: 'Ram',
    Address: 'Pune'
  },
  {
    _id: ObjectId('69eeb2bf0eb7daf5fce5f4a'),
    Roll_No: 4,
    Name: 'Sham',
    Address: 'Nashik'
  }
]

```

```

Stud> db.Stud.updateMany({Address:"Pune"},{$set:{Address:"Mumbai"}});
{
  acknowledged: true,

```

```

insertedId: null,
matchedCount: 2,
modifiedCount: 2,
upsertedCount: 0
}
Stud> db.Stud.find();
[
  {
    _id: ObjectId('69ee2fb2bf0eb7daf5fce5f48'),
    Roll_No: 2,
    Name: 'Sita',
    Address: 'Mumbai'
  },
  {
    _id: ObjectId('69ee2fb2bf0eb7daf5fce5f49'),
    Roll_No: 3,
    Name: 'Ram',
    Address: 'Mumbai'
  },
  {
    _id: ObjectId('69ee2fb2bf0eb7daf5fce5f4a'),
    Roll_No: 4,
    Name: 'Sham',
    Address: 'Nashik'
  }
]

},
{
  _id: ObjectId('69ee2fb2bf0eb7daf5fce5f49'),
  Roll_No: 3,
  Name: 'Ram',
  Address: 'Pune'
},
{
  _id: ObjectId('69ee2fb2bf0eb7daf5fce5f4a'),
  Roll_No: 4,
  Name: 'Sham',
  Address: 'Nashik'
}
]

```